7) $y' + 4y = 1.4$, $y = Ce^{-4x} + 0.35$, $y(0) = 2$

a)

$$\frac{d}{dx}(Ce^{-4x}) = -4Ce^{-4x}$$

$$(-4Ce^{-4x}) + 4(Ce^{-4x} + 0.35) = 1.4$$

$$-4Ce^{-4x} + 4Ce^{-4x} + 1.4 = 1.4$$

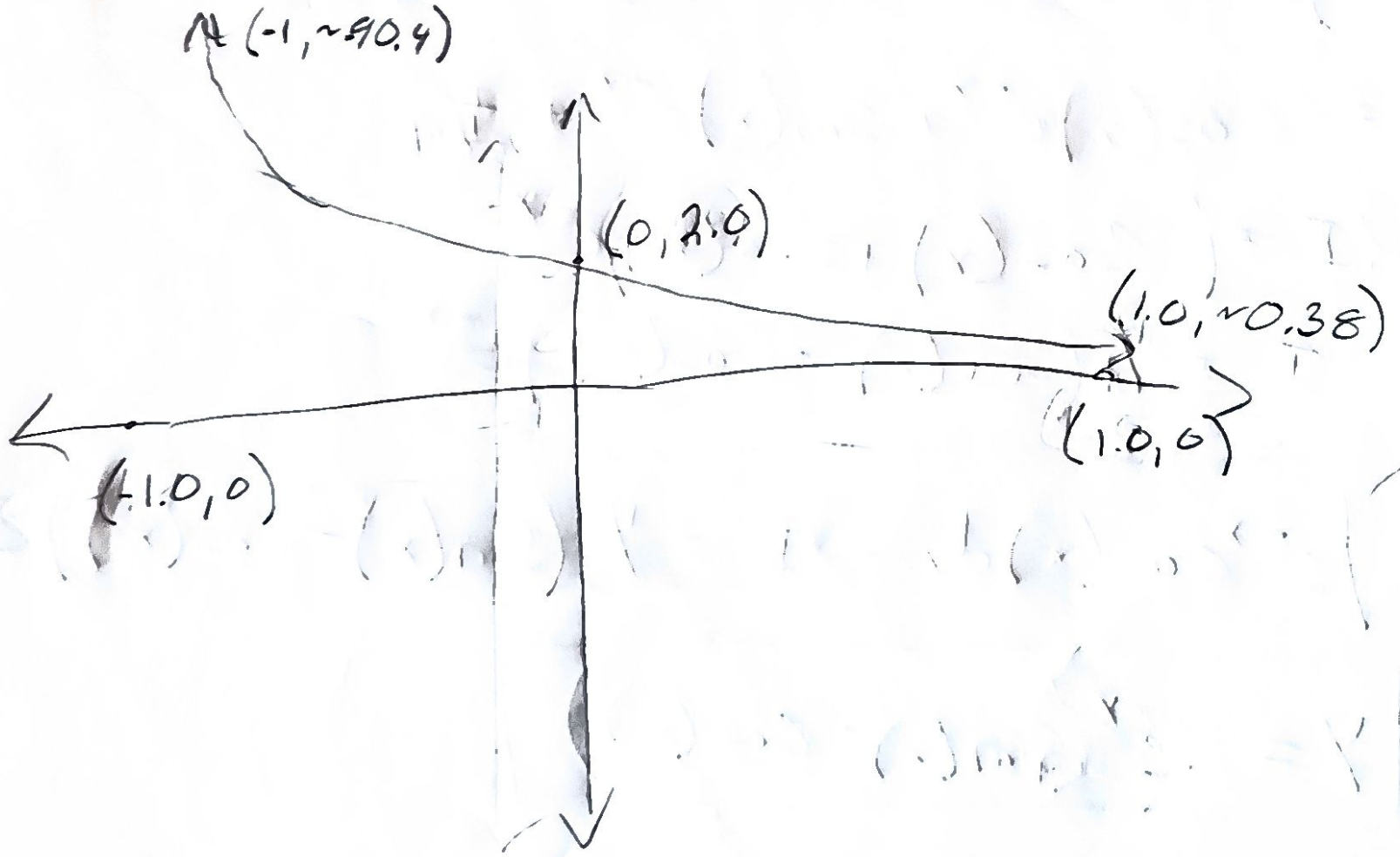
$$\boxed{1.4 = 1.4}$$

b) $y = Ce^{-4x} + 0.35 \Rightarrow 2 = Ce^{-4(0)} + 0.35$

$2 = C(1) + 0.35$

$C = 1.65 \Rightarrow \boxed{y = 1.65e^{-4x} + 0.35}$

c)



5] $y' = 4e^{-x}\cos(x)$ $\int u\,dv = uv - \int v\,du$

$4y = \int e^{-x}\cos(x)\,dx$ $u = \cos(x),\ dv = e^{-x}$

$= \int e^{-x}\cos(x)\,dx$ $du = -\sin(x)\ v = -e^{-x}$

$= \left[\cos(x)\cdot -e^{-x} - \int -e^{-x}(-\sin(x))\,dx\right]$

$= \left[-\cos(x)e^{-x} + \int e^{-x}(\sin(x))\,dx\right]$ $u = \sin x\ dv = e^{-x}$ $du = \cos x\ v = -e^{-x}$

$= \left[-\cos(x)e^{-x} + (-\sin x e^{-x} - \int -e^{-x}\cos x\,dx)\right]$

$= \left[-\cos(x)e^{-x} + \sin(x)e^{-x} - \int e^{-x}\cos(x)\,dx\right]$

let $I = \int e^{-x}\cos(x)\,dx$

$I = -\cos(x)e^{-x} + \sin(x)e^{-x} - I$

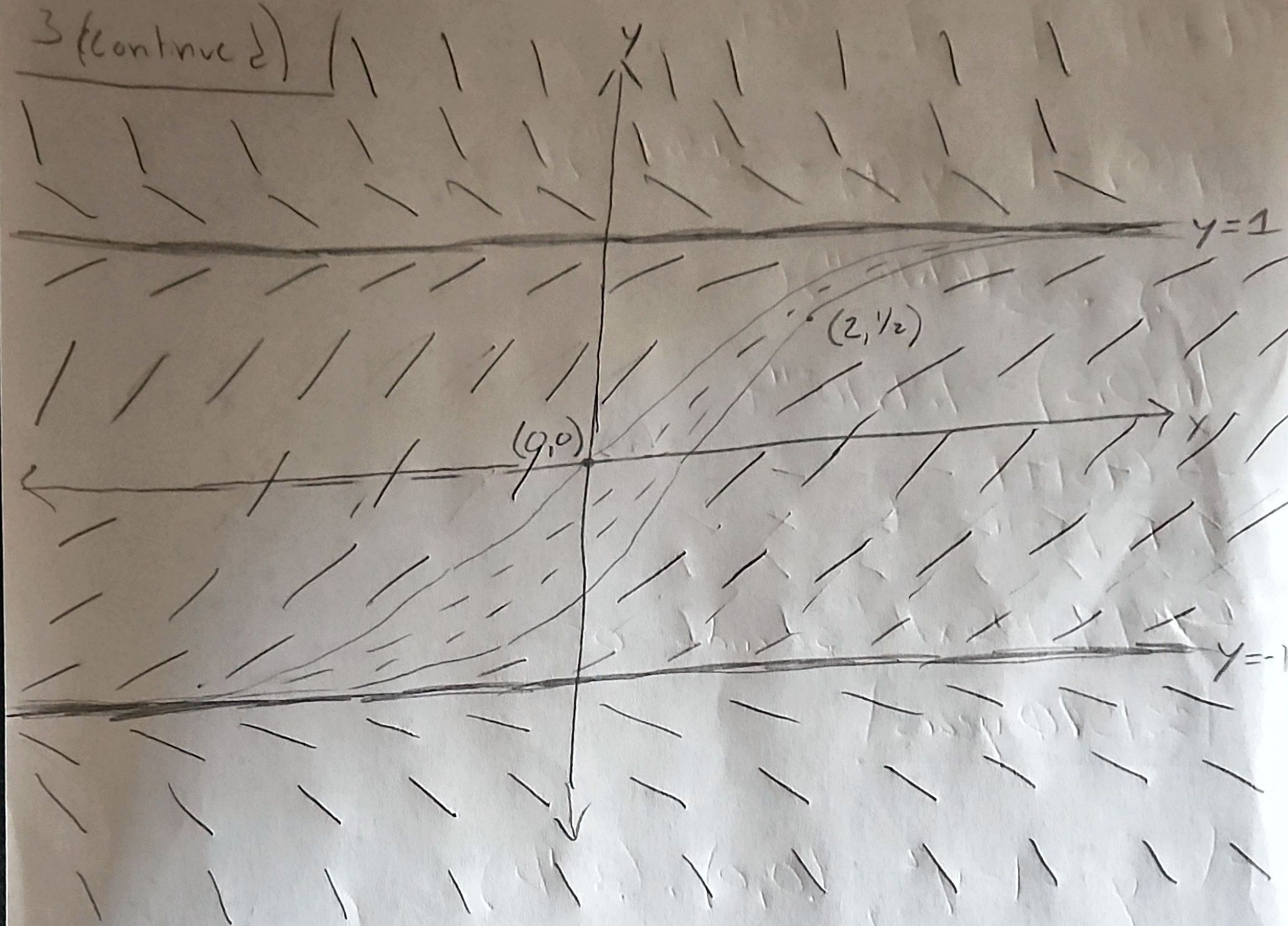
$2I = (-\cos(x) + \sin(x))e^{-x}$

$I = \frac{1}{2}(\sin(x) - \cos(x))e^{-x}$

$4y = \int e^{-x}\cos(x)\,dx = I = \frac{1}{2}(\sin(x) - \cos(x))e^{-x}$

$$\boxed{y = 2e^{-x}(\sin(x) - \cos(x))}$$

3 (continued)



17] Half life of $^{226}_{88}Ra$ in years.

$y(0) = 1, \quad y(t) = e^{-(1.4\times10^{-11})t}$

$0.5 = e^{-(1.4\times10^{-11})t}$

$\ln(0.5) = -1.4\times10^{-11}t$

$\frac{-0.693147}{-1.4\times10^{-11}} = t$

$t = 4.95105\times10^{10}$ seconds or

≈ 1570 years

3] $y' = 1 - y^2, \quad (0,0), (2, 1/2)$

Slope at $y = 0 \Rightarrow 1$
Slope at $y = 1/2 \Rightarrow 0.75$

$1 - y^2 = 0 = +/- 1$

15)

$y' = -4x/y$, $y(2) = 3$

$\frac{dy}{dx} = -4x/y$

$y\,dy = -4x\,dx$

$\int y\,dy = \int -4x\,dx$

$\frac{1}{2}y^2 = -2x^2 + C$

$\frac{1}{2}y^2 + 2x^2 = C$

$y^2 + 4x^2 = 2C$

evaluate at $y(2) = 3$ to find C

$(3)^2 + 4(2)^2 = 2C$

$9 + 16 = 2C$

$\frac{25}{2} = C$

$\frac{1}{2}y^2 + 2x^2 = \frac{25}{2}$

$y^2 + 4x^2 = 25$

$\boxed{y = \sqrt{25 - 4x^2}}$, for x where $y \neq 0$

19(Growth rate of bacteria that doubles/week

$dy/dt = ky$ where y is population at week, t

$y_0 = y(0)$, initial population

$dy/y = k\,dt$

$\int 1/y\,dy = \int k\,dt$

$\ln(y) = kt + C \rightarrow \ln(y_0) = C$

$y = e^{(kt+C)}$

$y = e^{kt}e^{C}$

$y = e^{kt}e^{\ln(y_0)}$

$y = y_0 e^{kt}$

Finding k

$y_0 = y(0)$ $\quad 2y_0 = y_0 e^{k}$

$2y_0 = y(1)$ $\quad 2 = e^{k}$

$k = \ln(2)$

$y = y_0 e^{\ln(2)t} \rightarrow \boxed{y = y_0 2^{t}}$